



Programming Skills Assessment

Data Structures & Algorithms — Coding Round

Each question carries marks as indicated. Read all questions carefully before starting.

Q1. Rotate an array by k positions to the right in O(1) space

Given an integer array, rotate the array to the right by k steps, where k is non-negative. You must do this in-place with O(1) extra space.

Time Complexity: O(n) **Space Complexity:** O(1) **Marks:** 1

Test Cases:

#	Input	Expected Output
1	[1, 2, 3, 4, 5, 6, 7], k=3	[5, 6, 7, 1, 2, 3, 4]
2	[1, 2], k=3	[2, 1]
3	[1], k=0	[1]
4	[1, 2, 3], k=1	[3, 1, 2]
5	[1, 2, 3, 4], k=4	[1, 2, 3, 4]

Q2. Find maximum product subarray (array can contain negatives)

Given an integer array, find the contiguous subarray (containing at least one number) which has the largest product and return its product. The array can contain positive numbers, negative numbers, and zeros.

Time Complexity: O(n) **Space Complexity:** O(1) **Marks:** 2

Test Cases:

#	Input	Expected Output
1	[2, 3, -2, 4]	6
2	[-2, 0, -1]	0
3	[-2, 3, -4]	24
4	[0, 2]	2
5	[-2, -3, -4]	12



Q3. Find if a number is a power of 2

Given an integer n , return true if it is a power of two. Otherwise, return false. An integer n is a power of two if there exists an integer x such that $n == 2^x$. Solve using bit manipulation.

Time Complexity: $O(1)$ **Space Complexity:** $O(1)$ **Marks:** 1

Test Cases:

#	Input	Expected Output
1	$n = 1$	True (2^0)
2	$n = 16$	True (2^4)
3	$n = 3$	False
4	$n = 0$	False
5	$n = -4$	False

Q4. Find the row with maximum 1s in a row-wise sorted binary matrix

Given an $n \times m$ binary matrix where each row is sorted in non-decreasing order, find the row with the maximum number of 1s. Return the row index (0-based). If two rows have equal 1s, return the one with smaller index.

Time Complexity: $O(n+m)$ **Space Complexity:** $O(1)$ **Marks:** 2

Test Cases:

#	Input	Expected Output
1	$[[0, 0, 1, 1], [0, 1, 1, 1], [0, 0, 0, 1]]$	1
2	$[[0, 0], [1, 1]]$	1
3	$[[1, 1, 1], [0, 0, 1], [0, 0, 0]]$	0
4	$[[0, 0, 0], [0, 0, 0]]$	0
5	$[[0, 1, 1, 1], [0, 0, 1, 1]]$	0

Q5. Next greater element for each item in an array

Given an array, for each element find the next greater element to its right. If no greater element exists, output -1. Return the results in an array. Solve using a monotonic stack in $O(n)$.

Time Complexity: $O(n)$ **Space Complexity:** $O(n)$ **Marks:** 2

Test Cases:

#	Input	Expected Output
1	$[4, 5, 2, 25]$	$[5, 25, 25, -1]$
2	$[13, 7, 6, 12]$	$[-1, 12, 12, -1]$
3	$[1, 2, 3, 4]$	$[2, 3, 4, -1]$
4	$[4, 3, 2, 1]$	$[-1, -1, -1, -1]$
5	$[5]$	$[-1]$

**Q6. Trapping rainwater — find total water trapped between bars**

Given n non-negative integers representing an elevation map where the width of each bar is 1, compute how much water it can trap after raining.

Time Complexity: $O(n)$ **Space Complexity:** $O(1)$ (two-pointer approach) **Marks: 2**

Test Cases:

#	Input	Expected Output
1	[0, 1, 0, 2, 1, 0, 1, 3, 2, 1, 2, 1]	6
2	[4, 2, 0, 3, 2, 5]	9
3	[3, 0, 3]	3
4	[1, 0, 1]	1
5	[0, 0, 0]	0

Total Marks: 10



Pattern Printing Questions

P13. Number border with alphabet fill

Print a square pattern of size n where the border is filled with the row number and the interior is filled concentrically with alphabets — A for the outermost inner ring, B for the next ring inward, and so on.

Marks: 1

Sample output (n = 7):

```
1      1      1      1      1      1      1
2      A      A      A      A      A      2
3      A      B      B      B      A      3
4      A      B      C      B      A      4
5      A      B      B      B      A      5
6      A      A      A      A      A      6
7      7      7      7      7      7      7
```

P14. Diamond — * border, alphabets from center outward

Print a diamond shape where the outermost shell is made of '*'. Inside the shell, 'A' sits at the top tip and the center of each row. Letters increase (B, C, D...) as you move outward from the center toward the * border. The pattern is symmetric both horizontally and vertically.

Marks: 1

Sample output (n = 5):

```
      *
    * A *
  * B A B *
* C B A B C *
* D C B A B C D *
  * C B A B C *
    * B A B *
      * A *
      *
```